Connecting Dots with Copilot

Solutioning the Erdős Unit Distance Problem

Author: Judi Evans
Co-Author: an instance of Microsoft Copilot

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Most people assume creativity means breaking patterns. But sometimes the real breakthrough comes from understanding which patterns matter. #Joseph Howlett's article "OpenAI Announces AI's biggest math breakthrough yet" on ChatGPT's solution to the Erdos Unit Distance Problem was an irresistible problem to tackle.

I enlisted Copilot to see what solution we could define to the Erdős Unit Distance Problem.

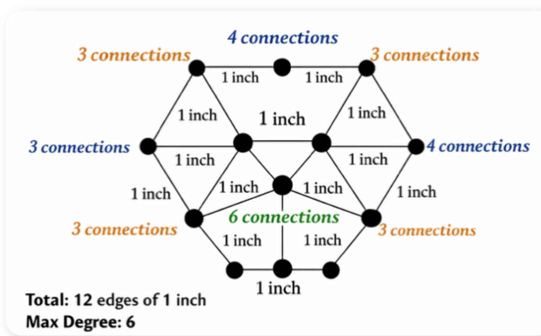
Place 9 dots on a sheet of paper.
How many pairs can you make exactly 1 inch apart?

I started with spirals, Copilot suggested lattices.

For 9 dots, under the exact 1-inch distance constraint, the determination was the 3×3 grid as the winner. It gives you 12 pairs of dots exactly 1 inch apart, and all 9 dots participate in at least one pair.

When you look at it from the perspective of a lattice, here's what happens with the same 9 dots:

- Square lattice: 12 one-inch pairs
- Triangular lattice: 15 one-inch pairs
- Hexagonal lattice: 18 one-inch pairs



Same number of dots. Different structure.
Completely different outcome.

The hexagonal lattice wins because it creates 2-dimensional adjacency. Each point becomes a hub, not a bead on a string. This is the same principle behind:

- network design
- identity governance graphs
- distributed systems

- organizational structures
- and even AI model architectures

The structure you choose determines the connections you can make. Creativity isn't about escaping structure. It's about choosing the right structure.

Sometimes the elegant spiral loses. Sometimes the boring grid wins. And sometimes the humble hexagon quietly outperforms them all.